Analysing Victoria Road Crash Data
Assignment 2:
BUSA8090 Data and Visualisation for Business

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1.Introduction:

This report analyses Victoria's road crash data to uncover key patterns in crash trends, severity, and vehicle involvement. Using R-based visualisations, it explores temporal, spatial, and vehicle-related factors influencing crash outcomes and demonstrates how ethical data visualisation supports evidence-based road safety insights.

2.Data Preparation, Transformation and Cleaning:

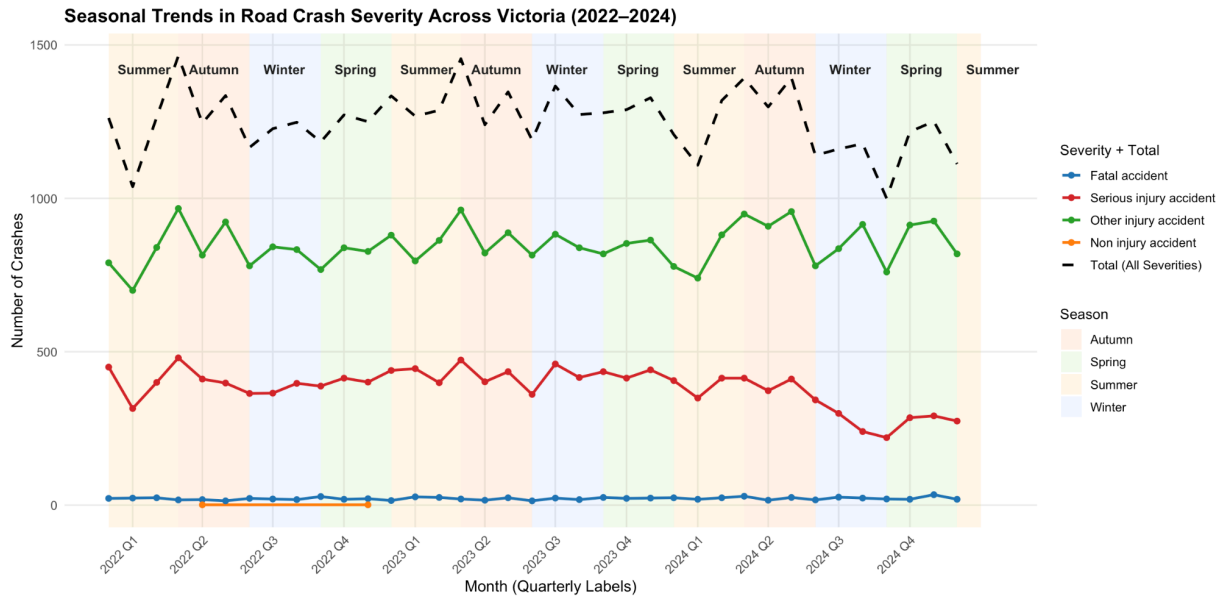
Exploratory Data Analysis (EDA) in Python initiated preparation for the three datasets, assessing missing values, formats, and variable types to identify data quality issues. Missing coordinates or inconsistent dates were reviewed and retained where still useful. This helped inform later cleaning in R.

Using tidyverse, lubridate, and ggplot2, the datasets (crash, vehicle, person) were merged by accident number. Dates were standardised, and coded variables such as crash severity and damage levels were replaced with descriptive labels. For the time-series visual, data was restricted to the latest three years, with seasonal groupings added for context.

Spatial and vehicle transformations used the full dataset. Records with valid coordinates were mapped for crash density, while vehicle data was grouped by manufacture year, type, and make to explore severity and damage patterns. This approach ensured a clean, consistent dataset for accurate visualisation.

3. Analysis of Road Crash Visualization:

3.1 Visualization 1: Seasonal Trends on Road Crash Severity Across Victoria (2022-2024)



Rationale for Choice:

A multi-line time-series chart with seasonal shading was used to show monthly crash counts by severity over three years. This format captures short-term fluctuations and broader seasonal cycles, allowing easy comparison between fatal, serious injury, other injury, and non-injury crashes. Seasonal colour bands (summer, autumn, winter, spring) provide context for interpreting environmental and behavioural influences on crash frequency.

Analysis of Findings and Interpretation:

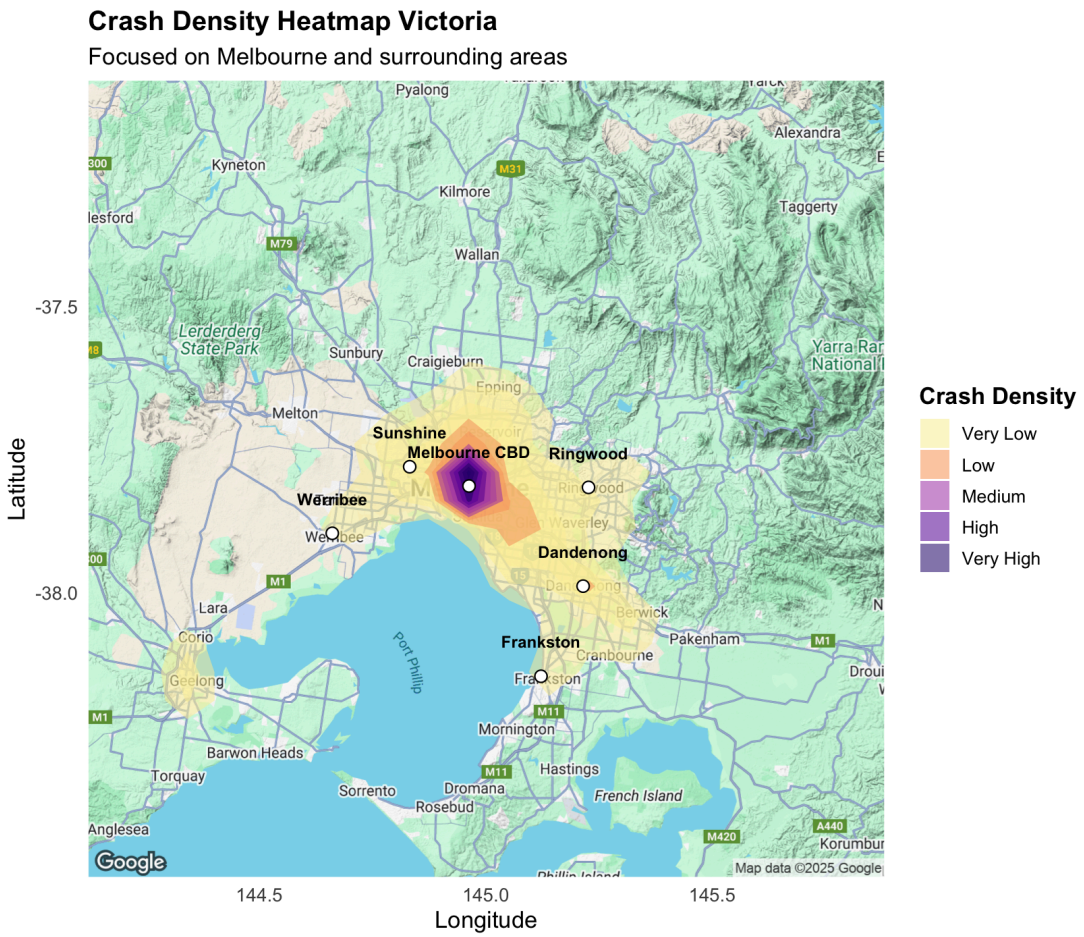
The visual shows other injury crashes (green) as the most frequent, followed by serious injury (red). Fatal crashes (blue) remain low and stable, while non-injury crashes (orange) appear minimal, likely due to reporting practices. A clear seasonal cycle emerges: crashes peak in summer and autumn, dip in winter, and rise again in spring. Increased travel during warmer months raises crash frequency, though not necessarily severity. The steady fatal crash rate indicates these are driven more by high-risk behaviours or isolated severe incidents than seasonal factors.

Potential Contributing Factors:

- I**ncreased mobility:** More travel for holidays and recreation in warmer months.

- **Weather and daylight:** Longer, clearer days encourage higher traffic and speed.
- **Holiday periods:** Festive seasons bring congestion, fatigue, and long trips.
- **Driver behaviour:** Risk-taking and distraction rise during holiday periods.

3.2 Visualization 2: Geographic Hotspots: Crash Density heatmap of Victoria



Rationale for Choice:

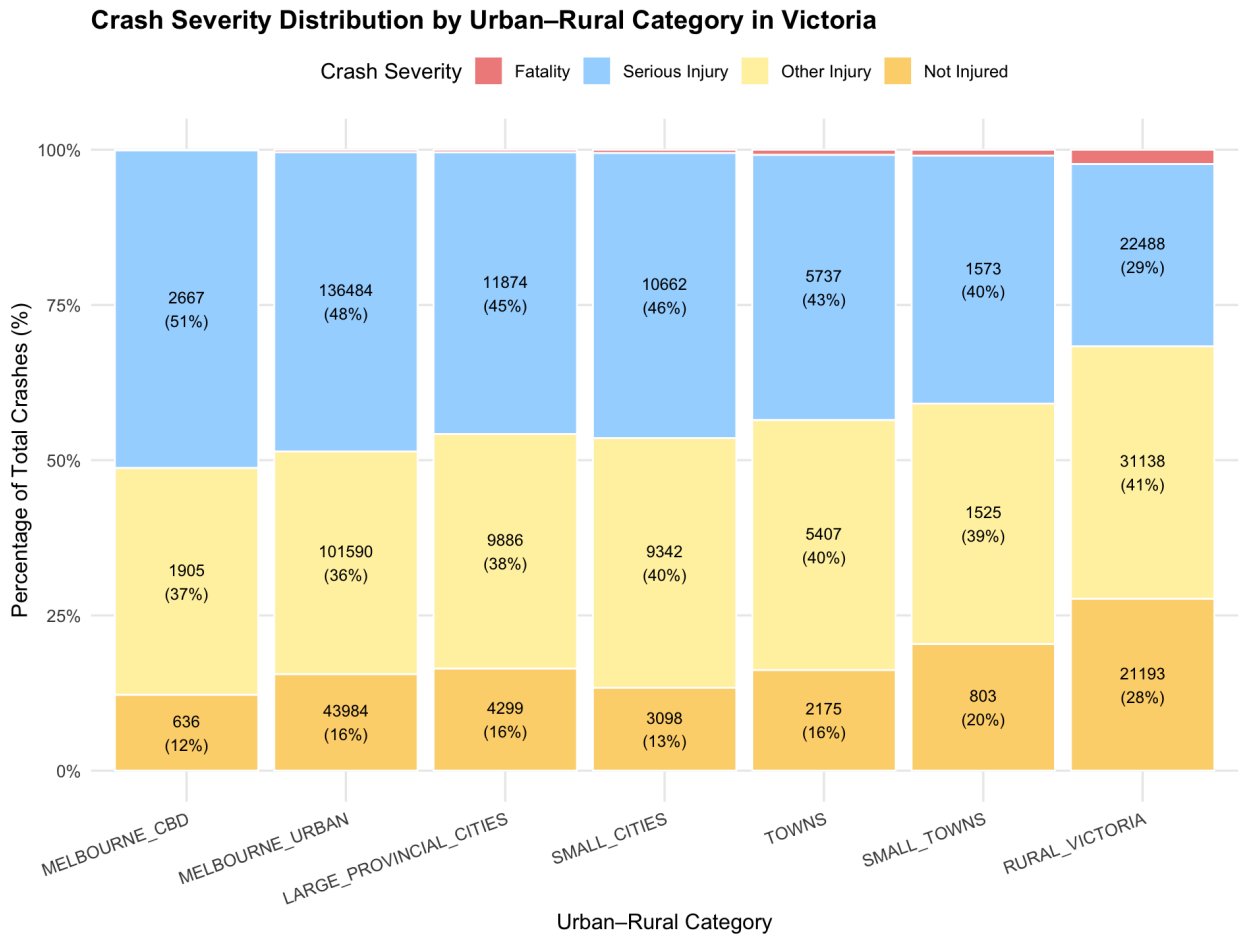
A heatmap was used to show the geographic distribution of crash density across Victoria, focusing on Melbourne and nearby suburbs. This visual highlights high-risk areas through colour intensity for spatial analysis. The terrain basemap adds context, while labelled suburbs such as *Melbourne CBD*, *Dandenong*, *Ringwood*, *Werribee*, *Sunshine*, and *Frankston* connect high-density zones to familiar locations.

Analysis of Findings and Interpretation:
The heatmap shows the Melbourne CBD as the main crash hotspot, with surrounding suburbs like Sunshine, Dandenong, and Frankston forming medium-to-high clusters. Density gradually decreases toward outer metro and rural areas. This pattern aligns with population and traffic distribution: urban centres have higher vehicle volumes, intersections, and complex road layouts, increasing collision risk. Rural regions show low density due to lighter traffic and simpler road networks. The clustering also follows major commuting routes and arterial corridors radiating from central Melbourne.

Potential Contributing Factors:

- **Urban traffic volumes:** High congestion and heavy vehicle flow raise crash exposure.
- **Intersection density:** Frequent turning and stop-start movement increase collision risk.
- **Commuting patterns:** Weekday travel and delivery activity heighten crash frequency.
- **Road complexity:** Urban areas have more intersections and pedestrian interaction than rural regions.

3.3 Visualization 3: Crash Severity distribution by Urban-Rural Category in Victoria



Rationale for Choice:

A stacked bar chart was used to compare crash severity proportions across seven urban–rural categories, from Melbourne CBD to rural Victoria. This design highlights how severity levels shift as road environments transition from dense urban areas to sparsely populated regions. Percentages were used instead of raw counts to standardise comparisons and prevent population size differences from distorting results.

Analysis of Findings and Interpretation:

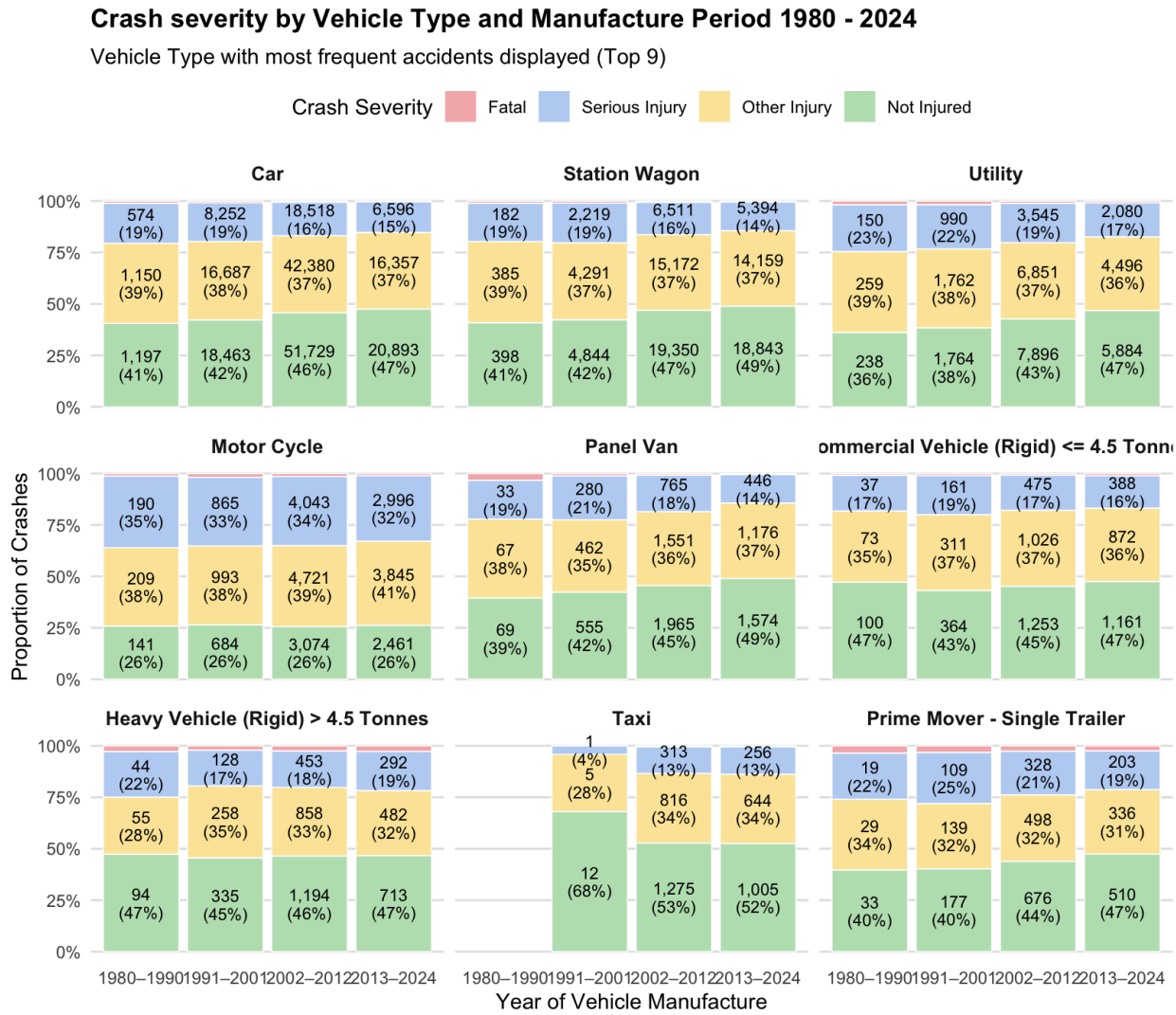
Melbourne CBD and Urban areas record most crashes but lower fatality rates. In contrast, rural Victoria experiences fewer crashes overall but a much higher share of fatal outcomes, indicating more severe impacts in high-speed zones. The share of serious injuries decreases in rural regions, suggesting that many crashes there result directly in fatalities. *Other injury* crashes (yellow) dominate across all categories, showing that most incidents lead to minor, non-hospitalised injuries. The *non-injury* category (orange) is more visible in small towns,

reflecting slower speeds and lower collision impact. Urban areas have frequent but less severe crashes, while rural regions face fewer but deadlier incidents

Potential Contributing Factors:

- **Speed environment:** Higher rural speed limits increase crash impact and fatality risk.
- **Emergency response times:** Longer delays reduce survival chances in severe crashes.
- **Road infrastructure:** Rural roads often lack lighting, barriers, or divided lanes.
- **Traffic density:** Urban congestion leads to minor collisions; rural crashes involve higher speeds and force.

3.4 Visualization 4: Crash Severity by vehicle type and manufacture period



Rationale for Choice:

A faceted stacked bar chart was used to compare crash severity across vehicle types and manufacturing periods from 1980 - 2024 as older vehicle before 1980 counts are extremely low. Each facet represents a vehicle category, allowing easy comparison between older and newer models. This format reveals how safety improvements over time influence crash outcomes. Percentages were used to focus on relative severity rather than total crash counts.

Analysis of Findings and Interpretation:

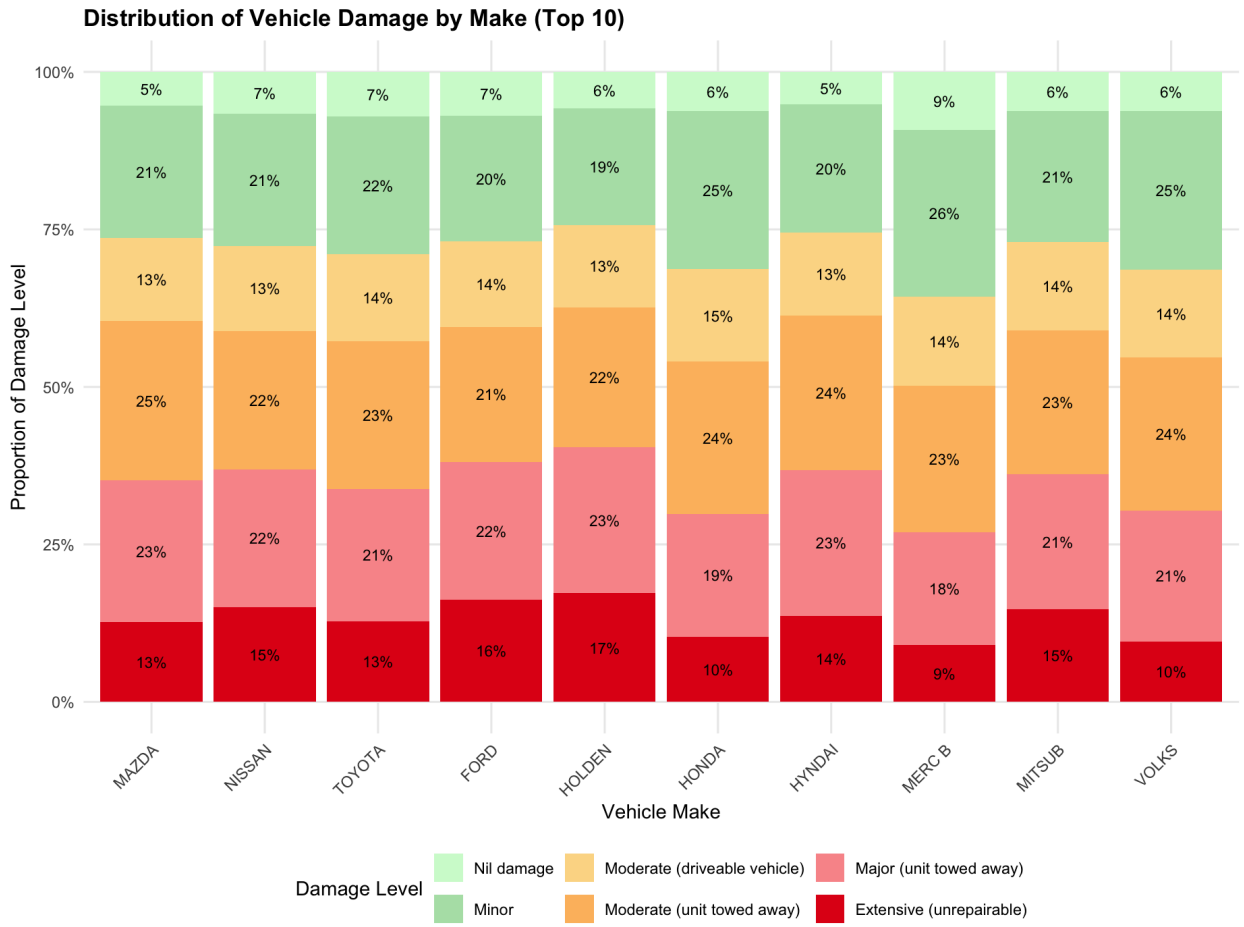
Newer vehicles (2013–2024) have higher non-injured outcomes, reflecting improved safety.. Cars, utilities, and station wagons built after 2012 record fewer fatal and serious injury crashes than older vehicles (1980–2001), reflecting the impact of features such as airbags, crumple

zones, and stability control systems. However, motorcycles display consistently high fatal and serious injury rates across all years, indicating the continued vulnerability of riders. Heavy vehicles also maintain moderate fatality levels due to the nature of large-mass collisions, while taxis show the lowest injury rates, likely because of frequent low-speed use and strict maintenance standards. Overall, newer designs reduce severity, though exposure and vehicle function still influence outcomes

Potential Contributing Factors:

- **Safety technology:** Modern systems like airbags and ESC reduce injury severity.
- **Vehicle exposure:** Heavy and commercial vehicles face higher mileage and crash risk.
- **Driver protection:** Motorcycles lack physical safety features, leading to higher severity.
- **Operating environment:** Taxis operate mainly in low-speed urban areas, limiting damage severity.

3.5 Visualization 5: Distribution of vehicle damage by make



Rationale for Choice

A stacked bar chart was used to compare vehicle damage levels among the ten most common makes involved in crashes. This format highlights how damage severity varies between brands while keeping proportions consistent. A colour gradient from light green (*nil damage*) to dark red (*extensive/unrepairable*) clearly communicates increasing severity, and percentages allow fair comparison despite differences in brand popularity.

Analysis of Findings and Interpretation:

The chart shows that Mazda, Toyota, and Nissan make up a large share of crash-involved vehicles but mainly experience *minor* or *moderate* damage. Holden and Ford display slightly higher shares of *major* and *extensive* damage, possibly reflecting older models and lower safety standards in ageing fleets. In contrast, Mercedes-Benz and Volkswagen have more *nil* and *minor* damage cases, suggesting stronger structures and modern safety features. Overall, damage patterns are fairly similar across brands, indicating no clear link between vehicle make and damage severity. Instead, external factors such as vehicle age, driving conditions, and

crash speed likely play greater roles. The dominance of minor damage suggests most crashes occur at lower speeds

Potential Contributing Factors:

- **Vehicle age and safety technology:** Newer models with advanced systems reduce severe damage.
- **Fleet composition:** Older vehicles (e.g., Holden, Ford) may be more vulnerable.
- **Build quality:** Premium makes use stronger materials and crumple zones.
- **Crash circumstances:** Urban low-speed impacts dominate most recorded crashes.

4. Ethical Considerations in Data Visualisation

Ethical data visualisation is vital for responsible analysis. This report adhered to key principles:

Accuracy and Honesty:
Data was presented truthfully and without distortion. Consistent scales, colours, and percentages were used across all visuals to maintain fairness. For example, the time-series chart showed actual monthly trends without exaggeration, and stacked bar charts used proportions to compare regions and vehicle types accurately.(Cairo, 2016)

Clarity and Avoidance of Misinterpretation:
Charts were designed to be clear and easy to understand. Appropriate types: such as line charts for trends, heatmaps for crash density, and stacked bars for severity and damage, were chosen. Clear labels and colour schemes made severity levels easy to interpret and prevented confusion. (Merson, 2018)

Transparency and Accountability:
Data sources and cleaning steps were clearly explained, including filtering invalid coordinates and grouping vehicle data by manufacture year. Only aggregated information was shown, protecting individual privacy while maintaining openness about methods and limitations.

5. Conclusion

The analysis of Victoria's road crash data highlighted strong spatial, temporal, and vehicle-related trends. Crashes peaked during summer and autumn, reflecting higher mobility and holiday travel. The Melbourne CBD and nearby suburbs showed the greatest crash density due to traffic volume and complex intersections, while rural regions experienced fewer but more fatal crashes linked to higher speeds and slower emergency responses. Vehicle analysis indicated that newer models with advanced safety technology reduced severe outcomes, and most crashes resulted in moderate or minor damage. No consistent link between vehicle make and severity was found, suggesting external factors such as speed and driving environment are more influential. Overall, these visualisations demonstrate how ethical, data-driven analysis can inform effective road safety strategies and support evidence-based policymaking in Victoria.

References:

Cairo, A. (2016). *The truthful art : data, charts, and maps for communication*. New Riders.

Merson, J. (2018). [Rev. of *Andy Kirk, Data visualization: A handbook for data driven design*]. *Environment and Planning. B, Urban Analytics and City Science*, *45*(2), 390–391. https://doi.org/10.1177/2399808317715320